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Assessment of water pollution of the Punpun River using biological parameters

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Abstract- India's urbanization and industrialization have made water pollution a serious problem. Monitoring biological parameters offers a quick, simple, understandable, economical, and time-efficient way to assess the overall effects of pollution. The river's biological factors must be evaluated in order to determine its current state because its properties are always changing. Faecal coliform levels at three locations in the Punpun River were studied, and the results showed notable seasonal changes. The pre-monsoon concentrations at the upstream location ranged from 2,700 to 8,600 MPN, suggesting significant contamination from agricultural runoff and local discharges. Concentrations rose during the monsoon because of increased river flow and runoff. As the river stabilized and sediment settled, post-monsoon levels dropped. Elevated pollution levels were suggested by the downstream site's regularly higher pre-monsoon faecal coliform values. The average faecal coliform concentration was 6,725.56 MPN for all sites, indicating variation in bacterial contamination. There were no discernible variations amongst the sampling stations, according to statistical analysis, indicating widespread effects from regional pollution sources.

Keywords: Punpun River, MPN, monsoon, faecal coliform, water pollution

INTRODUCTION

India's rivers, particularly those with religious significance, are heavily polluted by domestic sewage, industrial waste, and agricultural run-off. Major towns in river catchment areas discharge their wastewater into the rivers, posing environmental threats to humans and biodiversity. Non-point sources of pollution include agriculture runoff, cloth washing, and carcass disposal. The pollution potential has increased, affecting water quality, flora, and fauna. The Punpun River, a significant tributary of the Ganges, it extends for approximately 210 kilometers and is predominantly rain-fed, resulting in reduced water flow during the dry season. Originating in Gajandham, located in the Palamu District of Jharkhand, the river traverses several districts, including Chatra, Aurangabad, Gaya, and Patna, in both Jharkhand and Bihar. The river predominantly flows in a northeastern direction and converges with the Ganges at Fatuha, Patna. Throughout its course, the river has been contaminated by the careless disposal of household sewage, the submersion of ideals, bathing of cattle, runoff from agriculture and the massive amounts of partially treated and untreated sewage. The garbage from the settlements that line the river's banks enters into this as pollutants.[1]

Water pollution in India is a major issue due to industrialization and urbanization. Access to clean water is a basic human right, as stated in Article 21 of the Indian constitution. Polluted water is the root cause of many

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diseases. The state and citizens have a duty to raise nutrition, improve living standards, and protect the natural environment, including forests, lakes, rivers, and wildlife.[2] The parameters of water significantly influence the survival, growth, reproduction, and distribution of fish. Adverse environmental conditions can negatively impact aquatic life, as the interaction of various physical and chemical factors governs the health of aquatic ecosystems.[3] These parameters are crucial because they can directly or indirectly affect the quality, quantity, and diversity of fish and other aquatic organisms.

Biological Water Quality Criteria (BWQC) was introduced to determine the actual health of water bodies, for water quality evaluation.[4,5] Biological parameter monitoring provides an easy, effective, understanding, cost effective and less time-consuming method to determine collective impact of pollution.[6,7] As the characteristics of the river changes continuously it is necessary to assess its biological parameters in order to know its current situation.

MATERIAL & METHOD

Water samples for fecal coliform detection was collected in sterile containers to prevent contamination and analyzed as promptly as possible, within 6 hours of collection to ensure accuracy. For the analysis, Most Probable Number (MPN) method was performed to validate the presence of fecal coliforms. The concentration of fecal coliforms in the original water sample was calculated, expressed as colonies per 100 ml. All observations, including colony counts and confirmatory test results, was meticulously documented for quality control and reporting purposes.

RESULT

Table 1- Biological parameter status of Punpun River during sampling periods

Sampling periods	Sampling periods		Punpun ghat Road Bridge (Upstream)	Gaurichak bridge (Midstream)	Fathua Narayana Bridge (Downstream)
Fecal Coliform	Pre-monsoon	Minimum	2700	2650	2200
		Maximum	8600	8800	21000
		Average	5650	5725	11600
	Monsoon	Minimum	5200	4600	3800
		Maximum	7650	8910	18000
		Average	6425	6755	10900
	Post-monsoon	Minimum	1800	2100	3250
		Maximum	5800	6200	7800
		Average	3800	4150	5525

Table 2- Fecal coliform (One way ANOVA for comparing sites in all season)

Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	30893172.22	2	15446586.11	3.176583259	0.1145826506	5.14325285
Within Groups	29175850	6	4862641.667			
Total	60069022.22	8				

Table 3- Biological parameter status of Punpun River at all sampling sites during sampling periods

Parameter	Periods	N	Sampling sites			Mean	Statistics analysis	
			S1	S2	S3		std. Deviation	Std. Error Mean
Fecal Coliform	3	9	5291.6 7	5543.3 3	9341.6 7	6725.56	2269.11	1310.07

Table 4- Biological parameter status of Punpun River in seasonally of all site.

Parameters	Sampling Sites	Periods				Statistical Analysis	
		Pre-Monsoon	Monsoon	Post-Monsoon	Mean	Std. Deviation	Std. Error Mean
Fecal Coliform	3	7658.33	8026.67	4491.67	6725.5 6	1943.35	1121.99

FECAL COLIFORM

The study on fecal coliform concentrations at three sampling sites-upstream (Punpun Ghat), midstream (Gaurichak Bridge), and downstream (Fathua Narayana Bridge)-across pre-monsoon, monsoon, and post-monsoon seasons reveals significant variations in bacterial contamination levels, reflecting the impact of seasonal changes, hydrological processes, and anthropogenic activities on water quality. At the upstream site (Punpun Ghat), pre-monsoon fecal coliform concentrations ranged from 2,700 to 8,600 MPN, with an average of 5,650 MPN, indicating substantial bacterial contamination likely influenced by local discharges and agricultural runoff. During the monsoon, concentrations increased, ranging from 5,200 to 7,650 MPN, with an average of 6,425 MPN, attributed to runoff and higher river flow transporting contaminants. Post-monsoon levels decreased to a range of 1,800 to 5,800 MPN, averaging 3,800 MPN, likely due to the settling of sediments and contaminants as the river stabilized.

Similarly, at the midstream site (Gaurichak Bridge), pre-monsoon fecal coliform concentrations ranged from 2,650 to 8,800 MPN, with an average of 5,725 MPN, indicating comparable contamination sources to the upstream site. Monsoon concentrations rose to between 4,600 and 8,910 MPN, with an average of 6,755 MPN, mirroring upstream trends as runoff and increased flow contributed to bacterial levels. Post-monsoon concentrations decreased to a range of 2,100 to 6,200 MPN, averaging 4,150 MPN, as the river stabilized and contaminants settled.

At the downstream site (Fathua Narayana Bridge), pre-monsoon fecal coliform concentrations were significantly higher, ranging from 2,200 to 21,000 MPN, with an average of 11,600 MPN, suggesting that downstream areas experienced higher contamination, possibly due to cumulative pollution from upstream and local sources. Monsoon concentrations ranged from 3,800 to 18,000 MPN, with an average of 10,900 MPN, driven by agricultural and urban runoff, as well as increased contaminant transport by the river. Post-monsoon concentrations ranged from 3,250 to 7,800 MPN, with an average of 5,525 MPN, showing a reduction but still elevated levels compared to upstream and midstream sites, likely reflecting the persistence of cumulative contamination sources in the downstream region.

In this study, the overall fecal coliform concentration across all sampling sites of the Punpun River was found to have an average of 6,725.56 MPN with a standard deviation of 2,269.11, indicating variability in bacterial contamination across the river system. When analyzing the data seasonally, the average fecal coliform concentration was 6,725.56 MPN with a standard deviation of 1,943.35, demonstrating significant seasonal fluctuations in bacterial levels, particularly between the monsoon and non-monsoon periods.

The statistical analysis, including one-way ANOVA, was performed to determine if there were significant differences in fecal coliform levels between the different sampling sites. The results showed that there was no statistically significant difference in fecal coliform concentrations between the sites at the 0.05 significance level. This suggests that fecal contamination is widespread and relatively uniform across the upstream, midstream, and downstream sections of the river. The analysis of variance further confirmed the lack of significant differences between sampling stations, indicating that local pollution sources may be affecting the river system in a more general manner rather than contributing to distinct variations between specific sites.

DISCUSSION

Fecal coliform bacteria, a subset of total coliforms originating from the feces of warm-blooded animals, are crucial indicators of fecal contamination and potential pathogenic risks in water. Monitoring these bacteria is essential for assessing water quality, particularly for recreational and drinking water sources [8]. Fecal coliforms primarily enter river systems through agricultural runoff, wastewater discharge, and poorly managed septic systems, though wildlife and domestic animals also contribute. Elevated levels can pose significant health risks, including gastrointestinal and respiratory illnesses, and disrupt aquatic ecosystems by altering nutrient dynamics and affecting aquatic organisms [9,10].

In the Punpun River, the study of fecal coliform levels at three sites revealed significant seasonal variations. The upstream site showed pre-monsoon concentrations between 2,700 and 8,600 MPN, indicating substantial contamination from local discharges and agricultural runoff. During the monsoon, concentrations increased due to runoff and higher river flow. Post-monsoon levels decreased as sediment settled and the river stabilized. The

downstream site had consistently higher pre-monsoon fecal coliform concentrations, suggesting elevated contamination levels. The overall average fecal coliform concentration across all sites was 6,725.56 MPN, highlighting variability in bacterial contamination. Statistical analysis indicated no significant differences between sampling stations, suggesting general impacts from local pollution sources.

CONCLUSION

The investigation of faecal coliform levels at three locations in the Punpun River showed notable seasonal fluctuations. Pre-monsoon readings at the upstream location ranged from 2,700 to 8,600 MPN, suggesting significant contamination from nearby discharges and agricultural runoff. Because to runoff and increased river flow during the monsoon, concentrations rose. As silt settled and the river stabilised, post-monsoon levels dropped. Pre-monsoon faecal coliform values were consistently higher at the downstream location, indicating higher levels of pollution. There was variation in bacterial contamination, as seen by the 6,725.56 MPN average faecal coliform concentration across all sites. The lack of substantial variances between sampling stations, as revealed by statistical analysis, suggests that local pollution sources have a broad impact.

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CONFLICT OF INTEREST

None

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